

Projet statistique

Student Employability

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Chapter 1

PART ONE: Definition of Data and the Problematic

1.1 Definition and Formulation of the Problematic

The employability of students at ENSA of Kénitra is not limited to obtaining a diploma; rather, it results from a set of complementary factors that determine their professional integration within six months after completing their studies. Indeed, the relevance and quality of academic training, the alignment of technical and language skills with labor market needs, as well as the completion of concrete projects constitute key determining elements.

Added to this are the professional experiences acquired through internships and company-based projects, which strengthen practical understanding of the working world and enhance graduates' credibility with recruiters. Personal qualities—such as communication, adaptability, creativity, stress management, and teamwork—also

play a major role by complementing technical skills.

Finally, contextual factors such as the school’s reputation, labor market dynamics, and the strength of professional networks contribute to either facilitating or hindering access to a first job, making an in-depth analysis of all these dimensions essential for improving graduates’ employability.

1.2 Key Terms and Definitions

Key terms	Definition
Employability	Graduates’ ability to get a job within 6 months of finishing studies.
Professional Integration	Entering and settling into the job market after graduation.
Academic Training	The quality and relevance of the school’s formal education.
Technical & Language Skills	Job-market-ready hard skills in specific fields and languages.
Concrete Projects	Hands-on, practical work completed during studies.
Internships	Professional work experience gained in companies.
Personal Qualities	Soft skills like communication, teamwork, and adaptability.
Contextual Factors	External influences like school reputation and job market trends.
Professional Network	Connections that help in finding employment opportunities.

Table 1.1: Key terms related to student employability

1.3 Research Questions

1. **Academic factors:** To what extent does the quality of academic training (course content, mastery of technical skills, language proficiency, completed projects) influence graduates' employability within six months after obtaining their degree?
2. **Professional factors:** What is the impact of internships, company-based projects, and prior professional experiences on the speed and quality of students' professional integration?
3. **Personal factors:** How do transversal skills (soft skills), such as communication, teamwork, time management, and adaptability, contribute to graduates' employability?
4. **Contextual and environmental factors:** To what extent do professional networks, local opportunities, and the school's reputation influence graduates' access to a first job within a six-month period?

1.4 Documentary Research

The Documentary Research section provides a comprehensive review of existing literature on the employability and professional integration of engineering graduates, with a specific focus on the factors determining success in the labor market. It explores the complex interplay between academic training, acquired skills, professional experiences, and contextual elements, drawing on academic studies, institutional reports, and labor market analyses.

The review is structured to establish a theoretical foundation for the current study, which aims to analyze the determinants of employability for ENSA Kenitra

graduates. It systematically covers the definition and multidimensional nature of employability, the critical role of formal education and practical experience, and the importance of personal and contextual factors. Finally, it synthesizes key insights from the literature to frame the subsequent empirical investigation.

1.5 Data Dictionary

This data dictionary serves as a reference for structuring and standardizing the information collected from students as part of a study on their employability after completing their training program. The goal is to ensure uniformity, integrity, and traceability of the collected data to facilitate its analysis.

1.5.1 Entities

- **Student: Academic Background**

Attributes: Student ID, Age, Gender, Education Level, Major.

- **Professional Experience: Practical Background**

Attributes: Exp ID, Number of Internships, Internship Duration, Professional Network.

- **Personal Determination: Objectives and Motivation**

Attributes: Determination ID, Desired Profession, Motivation, Contract Type.

- **Skills and Characteristics: Aptitudes and Soft Skills**

Attributes: Skill ID, Technical Skills, Languages, Personality (Soft Skills).

1.5.2 Relationships Between Entities

- **Has_Training (1,1) to (1,1):** A student follows one main training program.

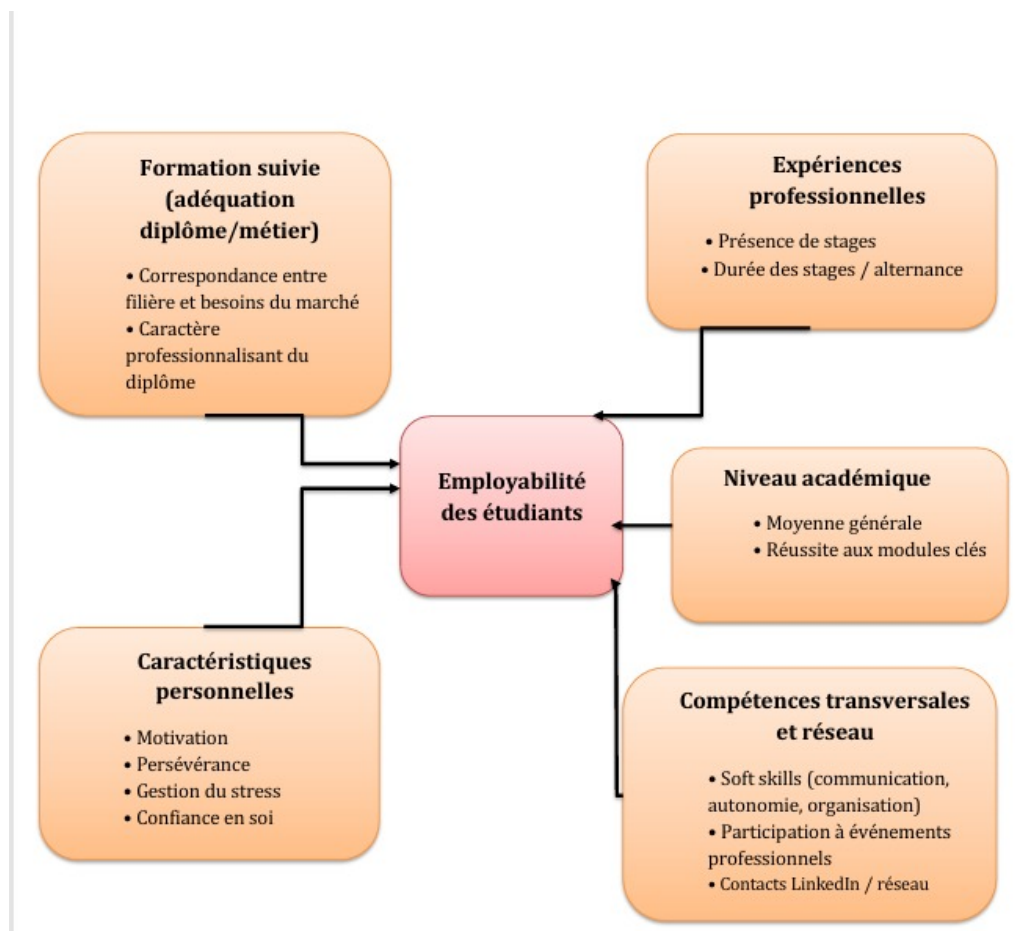
- **Has_Experience (1,1) to (1,1):** A student is described by their professional experience.
- **Has_Determination (1,1) to (1,1):** Personal determination linked to one student.
- **Master (1,N) to (0,N):** A student can master several skills.
- **Describe (1,N) to (0,N):** Training describes skills.

1.5.3 Variables Table

Variable Name	Type	Description	Constraint
Student_ID	Alphanum	Student identifier	Primary key
Age	Numeric	Student's age	≥ 16 and ≤ 27
Gender	Character	Student's gender	Male, Female
Education_Level	Character	Student's current level	1–3
Major	Character	Student's major	-
Exp_ID	Alphanum	Professional experience ID	Primary key
Internship_Count	Numeric	Number of internships	>0
Internship_Dur	Numeric	Duration in months	>0
Network	Numeric	LinkedIn connections	>0
Determ_ID	Alphanum	Determination ID	Primary key
Desired_Prof	Character	Profession desired	5–60 chars
Motivation	Character	Student's motivation	Salary, etc.
Contract_Type	Character	Desired contract	FTC, Perm...
Skill_ID	Alphanum	Skill ID	Primary key
Tech_Skills	Character	Software/Languages	5–100 chars
Personality	Character	Soft skills	5–100 chars
Language	Numeric	Number of languages	>0

Table 1.2: Data dictionary variables

1.6 Theoretical Conceptual Data Model



1.7 Hypotheses

1.7.1 Research Hypotheses

- **H1**: Impact of Professional Experiences.
- **H2**: Impact of Academic Level.
- **H3**: Impact of Soft Skills and Networking.
- **H4**: Impact of Personal Characteristics.
- **H5**: Impact of the Training Program (Degree/Job Fit).

1.7.2 Statistical hypotheses

H1: Professional experience influences employability.

- H_0 : There is no significant dependency between professional experience and employability.
- H_1 : There is a significant dependency between professional experience and employability.

H2: Academic level influences employability.

- H_0 : There is no significant dependency between academic level and employability.
- H_1 : There is a significant dependency between academic level and employability.

H3: Soft skills and professional network influence employability.

- H_0 : There is no significant dependency between soft skills/professional network and employability.
- H_1 : There is a significant dependency between soft skills/professional network and employability.

H4: Personal characteristics influence employability.

- H_0 : There is no significant dependency between personal characteristics (motivation, resilience, stress management) and employability.
- H_1 : There is a significant dependency between personal characteristics (motivation, resilience, stress management) and employability.

H5: Degree-job fit influences employability.

- H_0 : There is no significant dependency between the adequacy of the training program (degree-job fit) and employability.
- H_1 : There is a significant dependency between the adequacy of the training program (degree-job fit) and employability.

Chapter 2

PART TWO: QUALITATIVE STUDY

2.1 Introduction

As part of this statistics project, a qualitative study was conducted in order to gain a deeper understanding of the phenomenon under investigation beyond numerical data. Unlike quantitative analysis, which focuses on measuring and counting variables, qualitative research seeks to explore individuals' perceptions, behaviors, opinions, and experiences.

This qualitative approach makes it possible to analyze real-life situations, identify underlying factors, and better interpret the statistical results that will be obtained later. It provides essential context to the data, helps reveal trends, and gives meaning to numerical findings.

Through interviews, observations, and the analysis of participants' responses, this study aims to offer a comprehensive and detailed view of the subject. It also

helps in formulating hypotheses, guiding the quantitative analysis, and improving the overall quality of the project's conclusions.

Therefore, the qualitative study plays a key role in understanding the research topic and in developing relevant and reliable recommendations based on meaningful data.

2.2 Methodology

2.2.1 Interview Guide

The following questions serve as the foundation for our research, organized according to the 'funnel method/méthode d'entonnoir' and aligned with our conceptual framework.

2.2.2 PART 1: Introduction: Interview Objective, Confidentiality Note, and Thanks

Interview objective: Hello, we are a group of Industrial Engineering students from ENSAK. We are conducting a project on student employability, defined as the ability to find a job quickly after studies. The aim of this interview is to better understand your experience regarding this topic.

Confidentiality Note: We guarantee that this interview is anonymous, with a video recording that will be viewed only by the Professor. Your answers will remain confidential and will be used solely for our university work.

Thanks: We sincerely thank you for your participation, and for taking the time to answer our questions.

2.2.3 PART 2: Identification of the Interviewees

- Age
- Gender (Female, Male)
- Field of Study/Academic Discipline
- Year of Study

2.2.4 PART 3: Core of the Interview with Open-ended Questions

General vision of Employability

- **Q1:** For you, what does "being employable" mean as a student?
- **Q2:** In your opinion, what makes one student more employable than another?
- **Q3:** Can you give an example of a student you consider highly employable, and why?

Personal Competencies (Soft Skills)

- **Q4:** Which personal qualities (soft skills) most help a student find a job and subsequently succeed in the professional world?

University and Academic Training

- **Q5:** Do you think the training you are following at ENSAK really helps you become employable? Why?
- **Q6:** Which courses, projects, or subjects helped you most to develop useful skills for the labor market?

- **Q7:** Improvement suggestion: What is missing, in your opinion, in the current training offer to meet the requirements of the Industrial Engineering market?

Internships, and Extra-Academic Experiences

- **Q8:** How important are internships or professional experiences for employability?
- **Q9:** In your opinion, what types of experiences (volunteering, clubs, personal projects) help the most, beyond mandatory internships?
- **Q10:** Tell me about an experience that really taught you something useful for your professional future.

Analysis of Obstacles and Responsibility

- **Q11:** What difficulties does a student encounter most often when finding a job or a quality internship?
- **Q12:** Do these difficulties come mainly from the labor market, the university/training, or the students themselves?
- **Q13:** Can you recount a situation where you personally encountered a major difficulty while looking for an internship or a job?

Solutions and Recommendations

- **Q14:** What could concretely improve the employability of ENSAK students according to you?
- **Q15:** What specific role should the university play to further help its students?

- **Q16:** Have you ever seen an action, a workshop, or a practice in another university or engineering school that could be useful here?

Final Priority

- **Q17:** If there had to be only one thing, what is the absolute priority that our school should put in place starting tomorrow to better prepare students for their professional insertion?

Conclusion of the interview

We would like to sincerely thank you for the time you have dedicated to answering this questionnaire. Your contribution is essential to the success of our project and will allow us to formulate relevant and useful recommendations for all students in our establishment.

2.2.5 Population Characteristics

Defining the study population is critical for establishing the contextual validity of our analysis. This research targets students from the National School of Applied Sciences (ENSAK), specifically those enrolled in two key departments: Industrial Engineering and Computer Engineering (Informatics).

To allow for a multidimensional analysis of student employability, the following demographic and academic variables were selected:

- **Age:** Collected to analyze the maturity of students regarding the labor market.

- **Gender:** To observe if male and female students face different obstacles regarding insertion into the workforce.
- **Year of Study:** This factor is crucial to compare the perspectives of students at the beginning of their curriculum versus those nearing graduation who are actively seeking internships or employment.
- **Field of Study:** Identify the students' areas of study, such as Computer Science Engineering or Industrial Engineering, as these may influence their perceptions of employability and insertion into the professional world.

Chapter 3

Interview Responses on Student Employability

Table 3.1: Réponses des étudiants sur l'employabilité

Quest.	Student	Reformulated Response	Ideas
Q1	A. Chafii	To me, being employable means having what companies are looking for : good technical skills, practical experience, and the ability to adapt.	• Meeting corporate demands • Technical skills • Adaptability
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	J. Douae	Being employable means having skills, soft skills, and other competencies . It is not just about the diploma, but what you can do.	• Hard skills • Soft skills
	A. A. Is-maili	Being employable means having the necessary skills and confidence . It is the ability to persuade recruiters.	• Skills and confidence • Persuading recruiters
	B. M. Sagna	To me, being employable as a student is the aptitude or capacity to find a job or an internship .	• Capacity to find a job • Securing internships
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	O. Motasim	Being employable means knowing how to transform technical and academic skills into concrete solutions.	• Transforming skills • Adaptability
Q2	A. Chafii	I think it is a combination of three things: concrete projects , interpersonal skills, and genuine motivation.	• Projects • Motivation
	B. M. Sagna	What makes a student more employable is their skills and how to showcase them.	• Skill quantity • Self-promotion
	O. Motasim	A student is more employable when he combines technical skills with transversal qualities.	• Hard/soft skills • Rapid integration
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	J. Douae	As already mentioned, it is the skills and soft skills.	• Skill set
Q3	A. Chafii	Communication, teamwork, and adaptability. In IT, things change fast.	• Communication • Teamwork
	B. M. Sagna	The essential qualities are communication and selling oneself.	• Communication • Selling oneself
	J. Douae	Adaptability is the most important because technologies change.	• Adaptability
	O. Motassim	Essential qualities include communication, active listening, and emotional intelligence.	• Active listening • Stress management
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
Q4	A. Chaffi	My PFA (End-of-Studies Project) was decisive. I had to develop a full web application alone.	• PFA project • Autonomy
	B. M. Sagna	My internship in an accounting firm. I understood how a company works.	• Corporate understanding • Rigor
	J. Douae	An internship where I had to apply my theoretical knowledge.	• Theory vs Reality
	O. Motassim	Participating in a Hackathon. We had 48 hours to deliver a product.	• Hackathon • Pressure management
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
Q5	A. Chafii	It provides a good base , but it is not enough. You have to do a lot of self-learning.	• Good academic base • Self-learning
	B. M. Sagna	I would say 50/50 . We have a lot of theory, but we lack practice on tools.	• 50/50 balance • Lack of practice
	J. Douae	The theoretical base is good , but we need more practical projects.	• Need for practice
	O. Motasim	It gives a solid base , especially in logic and algorithms.	• Solid logic • Personal effort needed
Q6	A. Chafii	We need more practical workshops and less theory. Updating the curriculum is essential.	• More practice • Cloud/DevOps
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	B. M. Sagna	Creating more partnerships with companies for real-world projects.	• Partnerships • Seminars
	J. Douae	Reviewing the programs to remove obsolete modules and adding soft skills workshops.	• Remove obsolete • Soft skills
	O. Motassim	Focus on project-based learning from the first year.	• Project learning • Practice focus
Q7	A. Chafii	The main difficulty was the lack of experience requested by recruiters.	• Experience required • No feedback
	B. M. Sagna	Finding the first opportunity is the hardest. Companies want operational people immediately.	• Entry barrier
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	J. Douae	The competition is high . There are many graduates for few positions.	• High competition • Lack of network
	O. Motassim	Matching the specific technical stack of the company.	• Tech stack matching
Q8	A. Chafii	It is crucial . My internship was found via LinkedIn.	• Crucial role • LinkedIn
	B. M. Sagna	Very important . Recommendations from alumni or professors open doors.	• Recommendations • Alumni help
	J. Douae	I see that " who you know " is as important as "what you know".	• Vital importance
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	O. Motas-sim	It accelerates the process. Being recommended guarantees at least an interview.	• Interview guarantee
Q9	A. Chafii	Yes, absolutely. In IT, opportunities are concentrated in big cities.	• Total mobility
	B. M. Sagna	Yes, I am ready to relocate. Kenitra has limited offers compared to Casablanca.	• Relocation ready
	J. Douae	Yes, mobility is part of the engineer's life.	• Engineer lifestyle
	O. Motas-sim	Yes, no problem. I prioritize the quality of the mission.	• Priority on mission
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
Q10	A. Chafii	It is mandatory . School teaches the basics, but I learned React/Node.js on my own.	• Mandatory • Udemy/YouTube
	B. M. Sagna	Fundamental . Technology evolves too fast for the school curriculum.	• Fundamental • Curiosity
	J. Douae	It accounts for 70% of my real skills .	• 70% of skills
	O. Motasim	Very important. It shows the recruiter that you are passionate .	• Shows passion
	A. Chafii	Yes, managing events taught me stress management and teamwork .	• Stress management • Teamwork
Q11			
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	B. M. Sagna	Absolutely. I improved my communication .	• Communication
	J. Douae	It helped me gain confidence .	• Confidence
	O. Motassim	Yes, holding a responsibility position in a club proves your leadership .	• Leadership
Q12	A. Chaffi	English is non-negotiable for documentation and code.	• English for tech
	B. M. Sagna	Crucial . Many interviews are in English now.	• Crucial factor
	J. Douae	Essential. Most good tutorials are in English.	• Learning resource
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	O. Motas-sim	It is a barrier to entry if you don't master them.	• Entry barrier
Q13	A. Chafii	Saturated but full of potential. There is demand, but they only want the "best".	• Saturated yet potential
	B. M. Sagna	Competitive. The junior market is tough.	• Competitive
	J. Douae	It is demanding. The diploma is not enough anymore.	• Demanding
	O. Motas-sim	Open to those who are adaptable.	• Open to adaptability
	A. Chafii	Yes, cautious but optimistic. I trust my skills.	• Cautious optimism
Q14 Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	B. M. Sagna	Yes. I have prepared well.	• Optimistic
	J. Douae	Moderately. The economic context scares me a bit.	• Moderate
	O. Motassim	Very optimistic. Digital transformation is ongoing.	• Very optimistic
Q15	A. Chafi	Code every day and build a portfolio (GitHub) from year 1.	• Daily coding • GitHub early
	B. M. Sagna	Don't neglect soft skills. Join clubs, meet people.	• Soft skills • Networking
	J. Douae	Be curious. Explore technologies outside the curriculum.	• Curiosity • English
Suite page suivante...			

Table 3.1 – Suite de la page précédente

Quest.	Student	Reformulated Response	Ideas
	O. Motas-sim	Balance your life. Recruiters hire a person, not a robot.	• Life balance • Personality

THEMES	AD	IK	YB	NB	AM	Total
Theory vs Practice (lack of concrete)	X	X	X	X	X	5
Mastery of specific software		X	X	X	X	4
Fundamentals (Prog, Data)		X		X	X	3
Self-training & Certifications	X	X	X	X	X	5
Communication (written/oral)	X	X	X	X	X	5
Teamwork / Collaboration	X	X	X	X	X	5
Adaptability / Flexibility	X	X			X	3
Time & stress management		X		X	X	3
Honesty / Self-confidence			X	X		2
Leadership / Initiative		X		X		2
Mandatory internships	X	X	X	X	X	5
Team projects	X	X	X	X	X	5
Associative life / Clubs	X	X	X	X		4
Personal projects (GitHub)		X			X	2
Lack of network	X		X	X		3
Lack of experience	X	X		X	X	4
Saturated market	X	X	X	X		4
Absence of local network	X					1
Mobility			X			1
Networking with Alumni	X		X		X	3
Workshops (CV, Interview)		X	X	X		3
Module for Pro Insertion	X	X		X	X	4
Work-Study system	X			X	X	3
Mentorship System	X					1

The saturation level per theme is calculated as:

$$\text{Saturation Level (per theme)} = \frac{\text{Number of occurrences (repeated words) for the theme}}{\text{Total occurrences (repeated words) across all themes}}$$

The global saturation level is calculated as:

$$\text{Saturation Level} = \frac{79}{85} \approx 0.9294 \approx 93\%$$

Since it is greater than 70%, this justifies the sample size.

Chapter 4

PART THREE: Quantitative Study

4.1 Introduction

This quantitative study investigates the key factors determining the employability of ENSA Kenitra graduates. Moving beyond the diploma, it analyzes the measurable impact of academic training, practical experience (internships, projects), personal skills (communication, teamwork), and contextual elements (market dynamics, professional networks) on successful job acquisition within six months of graduation.

Through statistical analysis of survey data, this research aims to identify the most significant predictors of professional integration. The findings will provide actionable insights to enhance graduate employability strategies at the institutional level.

4.2 Methodology

4.2.1 Form Structure

The following sections present the captured screenshots of the questionnaire used in this study to assess the employability of ENSA Kenitra graduates. These screenshots highlight the specific questions and Likert scale used to gather responses.

4.2.2 Images of Questionnaires

The questionnaire consists of a series of questions displayed using a Likert scale. The following images show different parts of the questionnaire.



L'Employabilité des étudiants

Ce questionnaire vise à collecter des données essentielles sur les éléments qui façonnent votre employabilité et vos horizons de carrière en tant qu'étudiant(e) ingénieur(e) à l'ENSA Kénitra. Votre contribution est fondamentale pour affiner notre compréhension de ces dynamiques. Nous vous exprimons notre gratitude pour le temps précieux que vous y consacrerez. La confidentialité de vos réponses est garantie et leur usage sera strictement limité au cadre de cette recherche.

yousra.belghazi@uit.ac.ma [Changer de compte](#)



 Non partagé

* Indique une question obligatoire

Quel est votre âge ?

20

Quel est ton genre ? *

Figure 4.1: Questionnaire Part 1

Quel est ton genre ? *

☒ Femme

☐ Homme

Quelle est votre filière ? *

☐ GI

☒ GIND

Quel est votre niveau actuelle ? *

☒ 1ère année

☐ 2ème année

Suivant

Effacer le formulaire

N'envoyez jamais de mots de passe via Google Forms.

Figure 4.2: Questionnaire Part 2

Analyse des Compétences Essentielles pour l'Emploi Des Ingénieurs De l'ENSA Kénitra

Avez-vous effectué un stages ou une alternance durant votre formation ? *

☐ Oui

☒ Non

Si oui, combien de stages avez-vous effectués au total ? *

☒ 0

☐ 1

☐ 2

☐ Plus que 3

combien de temps a-t-il durée ? *

☒ 0 mois

☐ Un mois

☐ Moins d'un mois

☐ Plus d'un mois

Figure 4.3: Questionnaire Part 3

Pensez-vous que vos résultats académiques ont influencé votre recherche d'emploi ? *

☒ Oui

☐ Non

Que pensez-vous des opportunités d'employabilité pour un étudiant a ENSA de Kénitra ? *

	Pas du tout d'accord	Pas d'accord	Neutre	D'accord	Tout a fait d'accord
La formation académique que je reçois prépare efficacement au marché du travail.	☐	☒	☐	☐	☐
Les compétences transversales (communication, travail en équipe...) sont importantes pour trouver un emploi.	☐	☐	☐	☐	☒
Ma filière d'études offre suffisamment d'opportunités professionnelles.	☐	☐	☐	☒	☐

Figure 4.4: Questionnaire Part 4

Ma filière d'études offre suffisamment d'opportunités professionnelles.	☐	☐	☐	☒	☐
L'école propose un accompagnement satisfaisant pour l'insertion professionnelle	☐	☐	☐	☒	☐
Le réseau professionnel (contacts, clubs, associations) joue un rôle essentiel dans l'employabilité.	☐	☐	☐	☐	☒
Les étudiantes ont besoin de plus de formations pratiques pour être employables	☐	☐	☐	☐	☒

Considérez-vous que l'auto-formation contribue à améliorer votre employabilité ? *

☒ Oui

Figure 4.5: Questionnaire Part 5

formations
pratiques pour
être employables

Considérez-vous que l'auto-formation contribue à améliorer votre employabilité ? *

☒ Oui

☐ Non

Quel type de contrat préférez-vous pour votre futur poste ? *

☐ CDI (Contrat à Durée Indéterminée)

☒ CDD (Contrat à Durée Déterminée)

☐ Intérim (Travail temporaire)

☐ Freelance (Travail indépendant)

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Ce formulaire vous semble suspect ? [Signaler](#)

Google Forms

Figure 4.6: Questionnaire Part 6

4.3 Population and Sampling

The study population consists of 1980 students enrolled at ENSA Kenitra. The sampling method used is stratified random sampling to ensure representation across different majors and academic levels. The sample size was determined based on a confidence level of 95% and a margin of error of 5%, resulting in a final sample of 324 students.

4.3.1 Sampling Size Calculation

58 people filled out the form, knowing that the population size is 236, the Confidence Level is 90%, and the margin of error is 8%. According to the SurveyMonkey website, we obtain a quantitative sample size of 74.



The image shows a sample size calculation interface. It has three input fields at the top: 'Taille de la population' with the value 236, 'Niveau de confiance (%)' with a dropdown menu set to 90, and 'Marge d'erreur (%)' with the value 8. Below these fields, the result 'Taille de l'échantillon' is displayed in a large green font as 74.

Taille de la population	Niveau de confiance (%)	Marge d'erreur (%)
236	90	8

Taille de l'échantillon

74

Figure 4.7: Sample Size Calculation

4.4 Questionnaire Results (Charts)

The following charts illustrate the distribution of responses collected from the students.

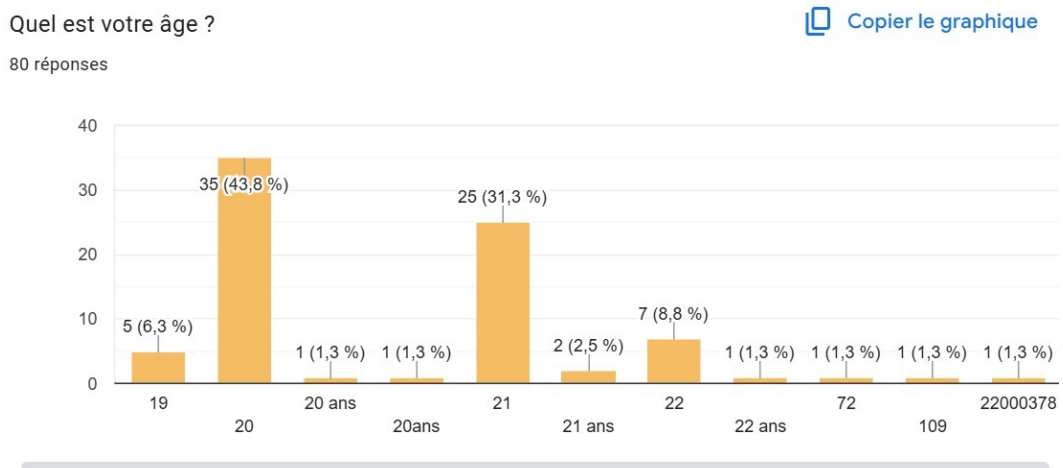


Figure 4.8: Result Analysis 1

Quel est ton genre ?
81 réponses

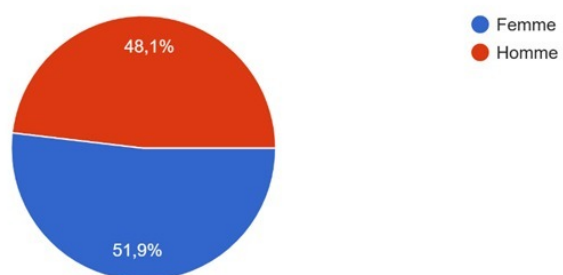


Figure 4.9: Result Analysis 2

Quelle est votre filière ?
81 réponses

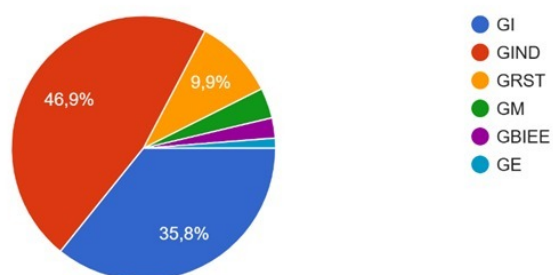


Figure 4.10: Result Analysis 3

Quel est votre niveau actuelle ?

81 réponses

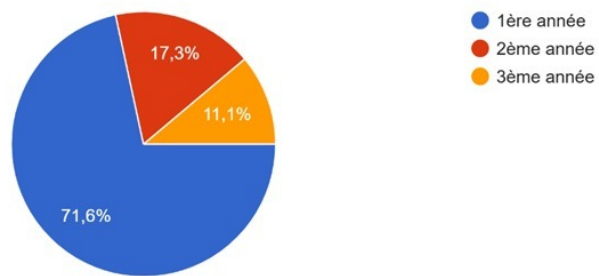


Figure 4.11: Result Analysis 4

Avez-vous effectué un stages ou une alternance durant votre formation ?
81 réponses

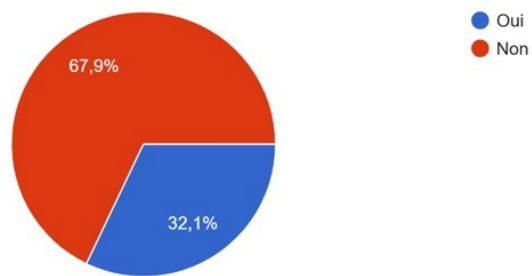


Figure 4.12: Result Analysis 5

Si oui, combien de stages avez-vous effectués au total ?
81 réponses

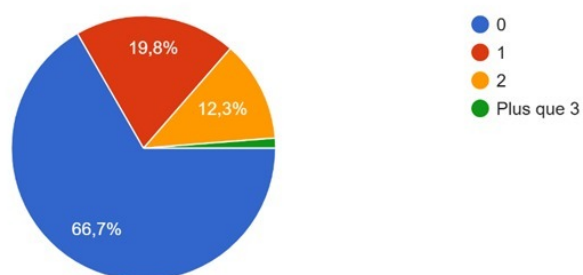


Figure 4.13: Result Analysis 6

combien de temps a-t-il durée ?
81 réponses

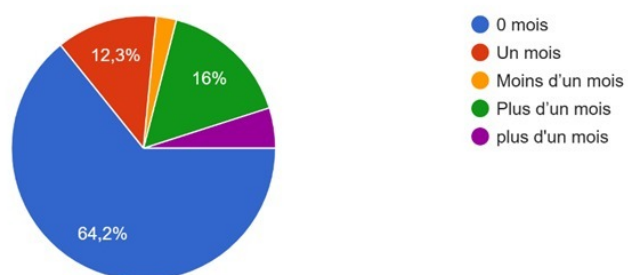


Figure 4.14: Result Analysis 7

Pensez-vous que vos résultats académiques ont influencé votre recherche d'emploi ?
81 réponses

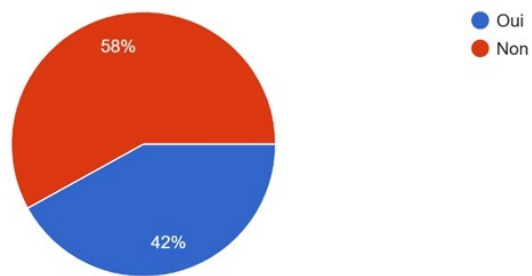


Figure 4.15: Result Analysis 8

Que pensez-vous des opportunités d'employabilité pour un étudiant a ENSA de Kénitra ?

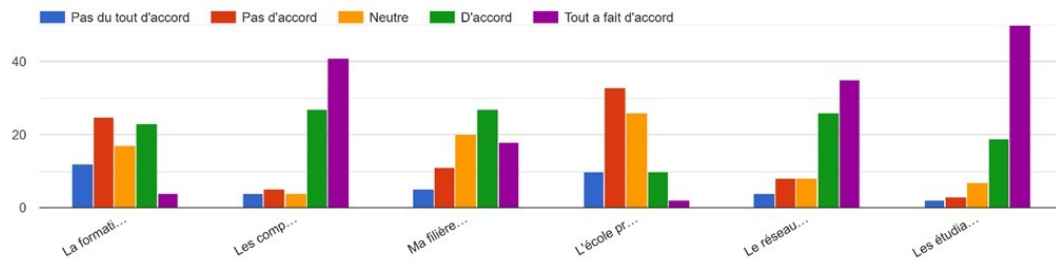


Figure 4.16: Result Analysis 9

Considérez-vous que l'auto-formation contribue a améliorer votre employabilité ?
81 réponses

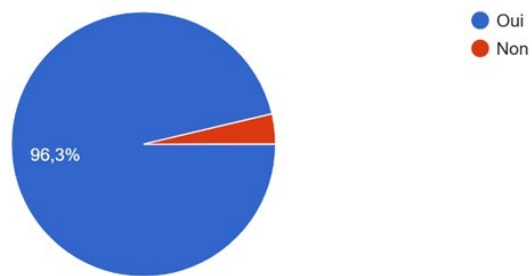


Figure 4.17: Result Analysis 10

4.5 Results and Hypothesis Testing

4.5.1 Hypothesis 1: Academic Training Status in the Job Market

The first hypothesis tested (H_1) addressed the suitability of academic training for the job market. The Chi-squared test was applied, yielding a p -value of 0.03352.

- **Interpretation:** Since the p -value is less than 0.05, we reject the null hypothesis (H_0). There is a statistically significant dependency between academic training and the job market, suggesting students perceive a significant gap.

4.5.2 Hypothesis 2: Importance of Transversal Skills

For hypothesis H_2 , we examined whether transversal skills are considered important for employment. The Chi-squared test produced an extremely low p -value of 1.292×10^{-9} .

- **Interpretation:** This very low p -value ($p < 0.05$) leads us to reject the null hypothesis. We conclude there is a highly statistically significant dependency, confirming that transversal skills are perceived as essential for finding a job.

4.5.3 Hypothesis 3: Professional Opportunities by Field of Study

The third hypothesis (H_3) investigated the link between the field of study and perceived professional opportunities. The Chi-squared test resulted in a p -value of 0.0003765.

- **Interpretation:** The p -value is less than 0.05, leading us to reject the null hypothesis. This indicates a significant dependency; the perception of professional opportunities varies significantly depending on the field of study.

4.5.4 Hypothesis 4: School Support for Professional Integration

Hypothesis H_4 assessed the satisfaction with school support for integration. The Chi-squared test produced a p -value of 8.488×10^{-7} .

- **Interpretation:** This p -value is well below 0.05, allowing us to reject the null hypothesis. There is a highly statistically significant dependency, suggesting a clear perception of the support provided by the school.

4.5.5 Hypothesis 5: Role of the Professional Network in Employability

For hypothesis H_5 , we explored the importance of the professional network. The Chi-squared test yielded an extremely low p -value of 5.271×10^{-8} .

- **Interpretation:** This p -value is very low ($p < 0.05$) and leads us to reject the null hypothesis. We conclude a highly statistically significant dependency, confirming the perceived essential role of the professional network in employability.

4.5.6 Hypothesis 6: Need for Practical Training for Employability

Hypothesis H_6 tested whether students perceive an increased need for practical training to be employable. The Chi-squared test resulted in an extremely low p -value of 7.584×10^{-14} .

- **Interpretation:** This extremely low p -value ($p < 0.05$) leads us to massively reject the null hypothesis. We conclude a highly statistically significant and extremely strong dependency, confirming the perception that there is a crucial need for practical training.

4.6 Quantitative Conclusion

In conclusion, our statistical analysis using R software is expected to reveal the key factors influencing student employability. Results from the survey and video interviews will allow us to validate or invalidate our initial hypotheses. We anticipate identifying significant correlations between, for example, the level of hard skills or soft skills and the professional integration rate. These findings will highlight current shortcomings and potential areas for improvement within our institution.

4.7 Decision-Making and Recommendations

This analysis aims to provide school decision-makers (administration, faculty) with clear perspectives on strategies to enhance overall student employability. The results obtained, based on the exploration of our Conceptual Data Model (CDM), will guide our recommendations. For instance, if the analysis highlights a lack of sector-specific

skills, we might recommend integrating new training modules or targeted practical workshops. The ultimate goal is to propose concrete actions to better align the academic curriculum with the demands of the job market.